

Control Co-Design: An engineering game changer

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Abstract

Over the last few decades, control engineers have focused on developing innovative control theories and algorithms to regulate systems. These control efforts are usually at the last stage of a sequential strategy that allows engineering departments to work independently and consecutively toward the design of new products and systems. Control algorithms are usually developed at the end of that sequential process, once the mechanical, electrical, and other subsystems are completely defined. This paper discusses a different approach, named *Control Co-Design* (CCD). Following a concurrent engineering strategy that considers multidisciplinary subsystem interactions from the beginning of the design process, CCD applies control concepts to design the entire system and reach optimal solutions that are not achievable otherwise. This approach is a *game changer* for the *control engineer*, who will be not only the designer of advanced control algorithms but also the natural leader of the design of new products and systems. This paper describes some historic engineering breakthroughs achieved by applying CCD and explores some relevant application areas. It also presents three complementary CCD methodologies that include control-inspired paradigms, formal mathematical co-optimization techniques, and co-simulation campaigns to enhance engineering creativity and achieve radically new optimal designs.

KEYWORDS

concurrent engineering, control, control co-design, design, systems engineering

1 | INTRODUCTION

Since the introduction of the very first servomechanisms long time ago to the current modern control solutions, control engineers have applied advanced mathematics, dynamics, physical models, algorithms, and technology to design autonomous systems. Establishing *feedback* as the central concept in the field, *Control Engineering* uses sensors to measure real-time data about system status, circuits and microprocessors to implement control algorithms, and actuators to modify automatically system variables and conditions. Paraphrasing Bode's seminal paper, it has been about *feedback, the [successful] history of an idea*.¹

Often, a control project starts when an engineering solution has been already proposed. That solution typically includes the definition of some mechanical, electrical, chemical, or biological subsystems, and some future control objectives. Based on that information, control engineers design and implement *control algorithms* that stabilize the system and achieve some reference tracking or disturbance rejection specifications. This control solution is typically developed at the final stage of a sequential design process; see Figure 1A.

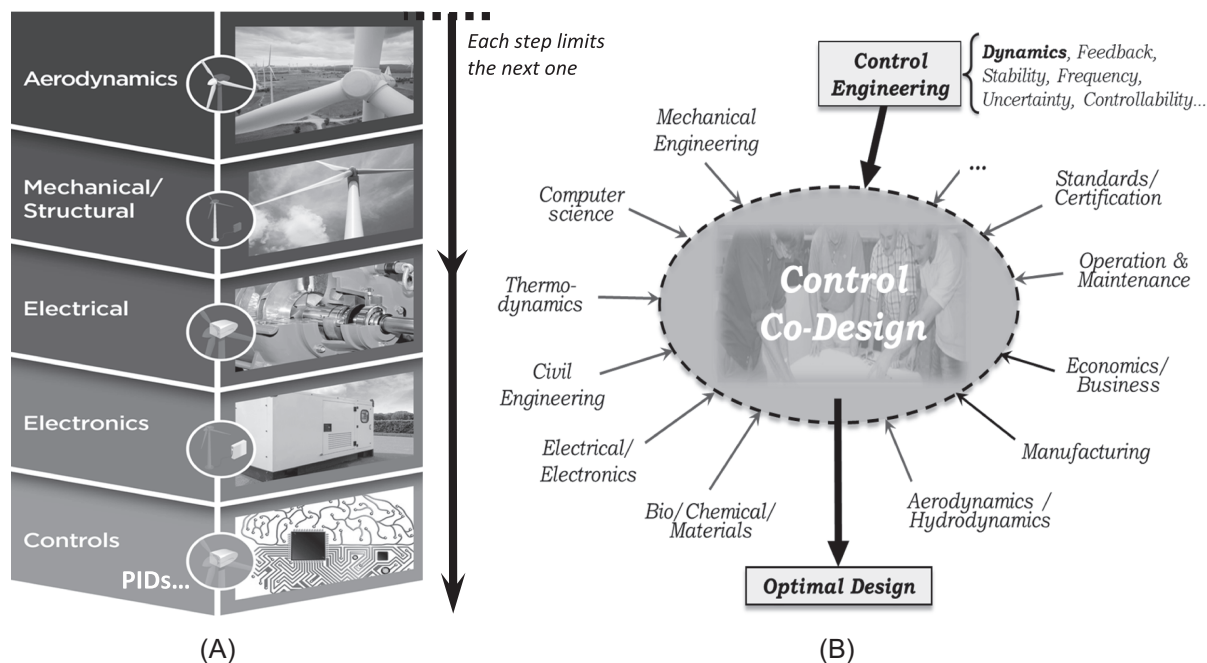


FIGURE 1 Sequential and Control Co-Design (CCD) approaches.² A, Sequential approach; B, CCD approach

As a result, control engineering research efforts have mainly focused on control theories and methodologies, from the popular proportional integral derivative (PID) controllers to more advanced robust, adaptive, optimal, or nonlinear control solutions.

On the other hand, the way we study engineering at college radically changed over the last few decades. As systems became more and more complex, engineering careers turned out much more specialized. Students and new engineers have today a deeper knowledge of some technical aspects at the cost of a much narrower picture.

Consequently, a sequential way of working of the engineering departments is now the common practice in industry; see Figure 1A. New products and systems are typically developed following a sequential approach, starting, for instance, with some mechanical engineering designs, continuing with aerodynamics/fluid discussions, then electrical/electronic solutions and at some point finishing up with the design and implementation of some control algorithms. An independent and sequential design process that develops the control system at a very late stage.

Although practical, this sequential approach limits the possibilities of the final products. Actually, every step of this process reduces the possibilities of the next step. In contrast, this was not the common approach in the late 19th century, before the specialization age. Great engineers of that period like Oliver Heaviside, Nikola Tesla, Wilbur and Orville Wright and Charles Brush kept a more complete and multidisciplinary picture, opening the possibilities of radically new and optimal designs. Giving the necessary credit to specialization, we need today to recover that multidisciplinary and concurrent engineering approach, especially when subsystem dynamic interactions are involved. This paper proposes this approach, named *Control Co-Design* (CCD), Integrated Design, or just Co-Design, which brings together all technical disciplines to work concurrently from the start and gives a special emphasis to the system dynamics and control paradigms; see Figure 1B.

Multidisciplinary systems might not be fully optimized unless subsystem interactions are considered. This optimization is particularly difficult when system dynamics are involved. CCD techniques consider these subsystem dynamic interactions from the very beginning of the design process. They use feedback, stability, frequency-domain characteristics and controllability concepts, and propose new mechanisms and control components to modify the system dynamics and reach optimal solutions that are not achievable otherwise. In summary, CCD enables a more optimal design with better system dynamics and controllability, which often results in lower system cost and improved reliability.

The individuals who can play a central role in this concurrent design approach are the *control engineers*. They apply dynamics and control concepts to understand the interactions among subsystems and can naturally lead all the

engineering disciplines to achieve a better system's dynamics, controllability and optimal design. This new role is a *game changer* for the *control engineer*, who will be not only the designer of advanced control algorithms, but also the natural leader of the design of new products and engineering systems in industry.

After this short introduction, this paper discusses some historical engineering achievements based on CCD breakthroughs, presents some CCD methodologies, and analyzes some key applications and future areas for the CCD approach.

2 | CCD PIONEERS

As said, great engineers of the late 19th century had a multidisciplinary understanding of systems that facilitated the design of radically new engineering achievements. This section discusses two historic engineering breakthroughs achieved by the Wright brothers and Charles Brush. They respectively designed the world's first heavier-than-air powered airplane and the world's first automatically operating wind turbine for electricity generation. In both cases, they unexpectedly proposed successful designs based on original CCD concepts.

2.1 | The Wright brothers

The Wright brothers, Wilbur (1867-1912) and Orville (1871-1948), designed the world's first successful, heavier-than-air, powered airplane. They made the first flight on December 17, 1903, four miles south of Kitty Hawk, North Carolina; see Figure 2 and author's other work.³

Flying was an impossible dream at the end of the 19th century. Many exceptional engineers preceded the Wrights and proposed a large variety of designs that eventually failed. The common understanding of *the flying problem* at that time focused on increasing the lift properties of the flying machines, often increasing the wings' surfaces or the number of wings, with biplane or triplane solutions. All these attempts to fly only considered *stable* airplanes. The stable-flyer concept was not even questioned. It was an intrinsic and needed characteristic of every design.

The Wrights, however, realized that the agility of an airplane to deal with wind disturbances was a key aspect and proposed a new revolutionary idea based on an *unstable* airplane.³ The new design incorporated a small wing at the front of the airplane; see Figure 2. This wing changed the relative position of the centers of gravity and pressure, being now the center of pressure closer to the front and the center of gravity closer to the rear of the plane. As a result, the airplane was inherently unstable.

That was a counter intuitive idea, especially at that time. In an undisturbed flight, the airplane naturally tended to turn upside down. However, this design increased the bandwidth of the dynamics and speeded up the airplane response to wind disturbances. Adding an appropriate longitudinal and lateral control system regulated by the pilot, the airplane became stabilized, keeping also an agile response for disturbance rejection.

As noted, the brothers applied a CCD approach to understand the subsystems' interactions, to coordinate different engineering disciplines, and to develop a better airplane's dynamics, controllability, and optimal design. Actually, the main claims included in the brothers' patents are about control concepts.

The Wright brothers had extraordinary mechanical engineering skills and a deep understanding of practical control concepts, learnt while working in their bicycle shop in Dayton, Ohio. They designed bicycles, which are essentially unstable systems controlled by a trained rider. Retrospectively, we can recognize them as truly pioneers of CCD.

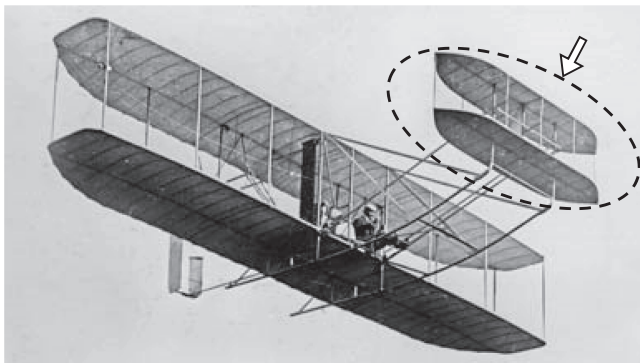


FIGURE 2 Wright brothers' unstable airplane. December 17, 1903: the world's first heavier-than-air powered and controlled human flight (US Library of Congress)



FIGURE 3 Charles Brush's wind turbine, 1887. The world's first automatically operating wind turbine for electricity generation (Kelvin Smith Library, CWRU, and Scientific American)

2.2 | Charles Brush

Charles F. Brush (1849-1929) is one of the founders of the electrical industry in the US. He invented DC dynamos, commercial arc-lights, and lead-acid batteries. He also designed the world's first successful automatically operating wind turbine for electricity generation. The turbine was designed and erected in Cleveland, Ohio, in 1887, and was in operation until 1908; see Figure 3 and author's other work.³

The wind turbine was a giant, the world's largest, with a rotor diameter of 17 m and 144 blades made of cedar wood. At full load and 500 rpm, the electrical generator power output was 12 kW—one rotor turn corresponded with 50 revolutions of the generator. It was used to supply energy through underground conductors to 408 battery cells, 350 incandescent lights, 2 arc-lights, and 3 electric motors in Brush's mansion on 37th Euclid Avenue in Cleveland.

In July 1887, an enormous summer storm lashed out northeast Ohio, destroying the wind turbine, which had become a very popular and unique structure for the Clevelanders, according to a local newspaper, the Plain Dealer.

In just one year, Charles Brush rebuilt the turbine. Instead of reinforcing the tower and blades to survive potential future extreme weather conditions, Brush applied a CCD approach.³ He designed a new yaw control system composed of a large tail (18 × 6 m), an auxiliary vane perpendicular at one side, and a mechanism of weights and pulleys inside the tower; see Figure 3. Under normal wind conditions, the large tail of the yaw control system was able to point the turbine rotor perpendicular to the wind direction for maximum energy generation and efficiency. Eventually, under high winds, the auxiliary vane became dominant over the internal weights, being able to turn the rotor out of the wind direction, reducing the mechanical loads and rotor velocity. In other words, it was a radically new design that proposed a control system as a substitute of additional materials—a truly CCD solution.

The turbine had also what it is probably the world's first automatic DC voltage regulator: a group of automatic switching control devices arranged so that the dynamo was into effective action at 330 rpm, and the DC voltage at the batteries was kept between 75 and 90 V.

Charles Brush described the wind turbine as quote “self-regulating, requiring no person to attend it.” The improved turbine operated for the next 20 years without any problem, delivering 12 kW of power to Brush's home until the decommission in 1908. Retrospectively, we can also recognize Charles Brush as a truly pioneer of CCD.

3 | APPLICATIONS AND RECENT WORK

With a concurrent control engineering approach, the CCD methodology opens numerous opportunities for optimal and radically new designs. This section explores some promising engineering areas that can significantly benefit from this methodology. In general, CCD plays a fundamental role in finding an optimal design when there exist a dynamic interaction between the subsystems that comprise the plant. In other words, CCD is not relevant when the subsystems and components are independent from each other. However, it is vital when the internal subsystems are dynamically coupled. The higher the subsystem dynamic interactions, the more decisive the CCD methodology.

Some CCD formal discussions started at the end of the 1980s, with an inspiring proposition that emphasized the simultaneous co-optimization of the parameters of the controller and the plant, both with a fixed structure.^{4,5} The new ideas

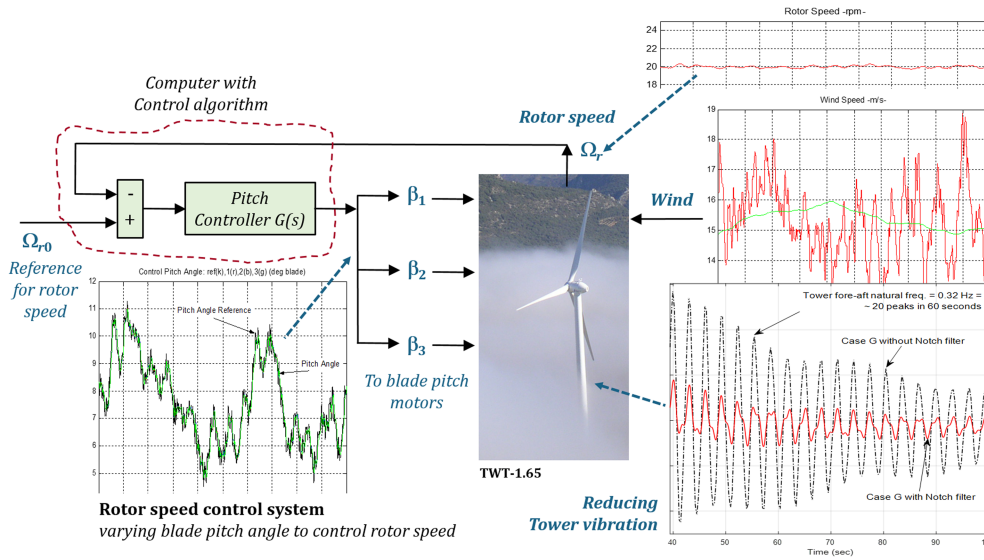


FIGURE 4 Control Co-Design of a commercial 1.65 MW direct-drive wind turbine^{3,18}

were soon known as integrated structure/control design and were first applied to airplane structures, flexible robots, and spacecraft.⁴⁻⁸ They considered the inherent controllability of the plant in the design of the system, including concepts as loop coupling, disturbance rejection, reference tracking, actuator constraints, and actuator and sensor location along with the mechanical structure of the plant.

Other research groups and institutions promptly followed these initial discussions, using these CCD ideas to design chemical reactors^{9,10} and aircraft.¹¹ Since the mid-1990s up to the present, the CCD methodology has been applied to numerous engineering areas, including underwater vehicles,^{12,13} networked control systems,¹⁴ wastewater treatment plants,¹⁵ automotive suspensions,^{16,17} wind turbines,^{3,18} bicycles,¹⁹ motorcycles,²⁰ composite smart antennas and mechanical structures,²¹ flexible space structures,²² airborne wind energy systems,²³ grid and power systems,²⁴ very flexible actuated structures,²⁵ variable stiffness actuators and robotics,²⁶ communication networks for intelligent vehicles,²⁷ automotive motor-transmission powertrains,²⁸ floating offshore wind turbines,^{2,29-31} and tidal and wave energy converters.^{29,30} Other relevant potential areas are biological and medical systems, digesters, water desalination, energy storage, conventional and renewable power generation, and power electronics.

Figure 4 presents an illustrative CCD example in which the author was involved at the end of the 1990s; see Case Study 2 in the work of Garcia-Sanz.^{3,18} The system was composed of a direct-drive, variable-speed, pitch-controlled 1.65 MW wind turbine. The machine, a type-4 turbine, did not need a gearbox and incorporated a full-power converter to control the aerodynamic efficiency and the grid variables simultaneously and independently. By applying CCD concepts to the pitch control system, the turbine achieved not only a smooth and robust rotor speed control but also a significant reduction of the tower vibration and the corresponding mechanical fatigue. This fatigue reduction allowed the company to design a machine with a tower considerably cheaper than the immediate competitor, with also better reliability and maintenance characteristics.

Starting in January 2018, the Advanced Research Projects Agency for Energy (ARPA-E), at the US Department of Energy, began challenging the research and industrial communities to develop new and disruptive CCD solutions for a large variety of applications.^{2,29-31} Subsequent efforts also included new activities by the National Science Foundation (NSF) and the Water Technology Office (WPTO) at the US Department of Energy.

The ATLANTIS Program, introduced by ARPA-E in 2018/19, proposed a CCD approach to find radically new designs of floating offshore wind turbines (FOWT).² The highly coupled dynamics involved in a FOWT make it an ideal candidate for the CCD approach. Figure 5 shows the main subsystems of a FOWT, including the rotor, drive-train, electrical generator, power electronics, substation, nacelle, tower, floating platform, mooring system, aerodynamics, hydrodynamics, grid, and control systems. The Figure also shows the main external inputs that affect the system, including wind, waves, tides, ocean currents, and grid voltage and frequency, and emphasizes the strong interaction of the aerodynamics, hydrodynamics and mechanical structure.

As an example of these interactions, let us consider some wind that moves the rotor of the turbine at a given rotational speed Ω_r ; see Figure 5. This rotor typically sweeps an enormous area and contains a very large moment of inertia I_r , especially in multi-megawatt systems. The rotational speed and moment of inertia define the angular momentum $I_r \Omega_r$.

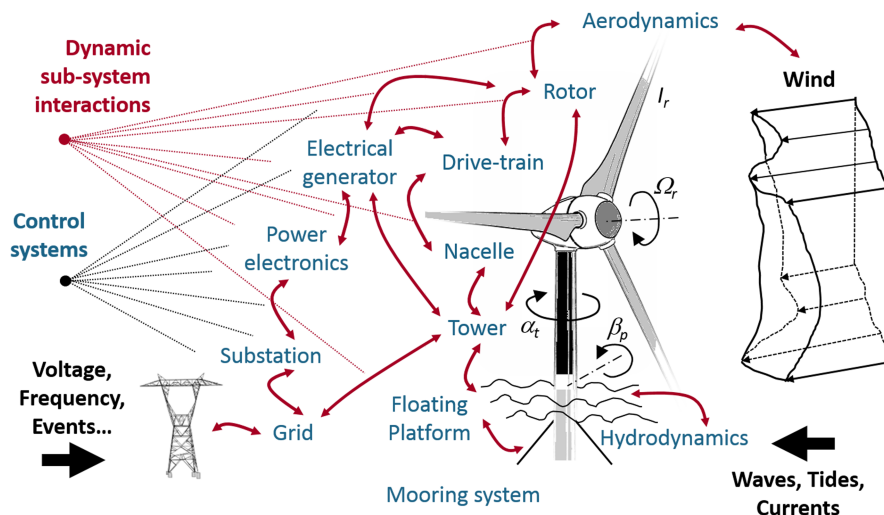


FIGURE 5 Subsystems' interaction in a floating offshore wind turbine²

Let us also consider that, with the turbine working at that rotational speed, a wave hits the system and changes the pitch angle of the floating platform β_p ; see also Figure 5. As a result, a gyroscopic effect will rotate the platform about an axis perpendicular to both the angular momentum $I_r \Omega_r$ and the angle β_p , moving the yaw angle α_t of the floating platform to keep the angular momentum constant—law of conservation of angular momentum.

This inherent interaction between the aerodynamic components (rotor), hydrodynamic components (floating platform) and intermediate mechanical structure (tower) illustrates the need of a CCD approach to optimize the FOWT system. Independent designs of wind turbine and floating platform cannot achieve an optimal solution.

FOWTs have many other important interactions between aerodynamics, hydrodynamics, mechanical structure, mooring system, electrical systems, and control systems. The analysis of all these subsystem interactions, the design of innovative components and control solutions to deal with those interactions, and the concurrent control engineering approach, or CCD, are critical to find optimal solutions.²

4 | CCD AREAS AND METHODOLOGIES

Over the last few decades, engineers and scientists have proposed a number of methodologies to develop CCD projects. Based on the complementary fields of Engineering, Mathematics, and Computer Science, these methodologies matured into three primary areas, which respectively are (A1) *control-inspired paradigms*, (A2) *co-optimization techniques*, and (A3) *co-simulation methods*. Figure 6 shows these CCD areas. In addition, every CCD project typically needs five fundamental inputs: (i1) *system objectives*, (i2) *components pre-design*, (i3) *physics-based models*, (i4) *real-data* from lab or full-scale experiments, and (i5) *case studies* to conduct the analysis.

The first input of the CCD process, (i1) *system objectives*, defines the goals and targets to be achieved by the final system. Usually, these objectives are formally defined in terms of numerical *metrics* that include a variety of characteristics like performance, dynamics, controllability, efficiency, robustness, survivability, lifespan, resiliency, economics, and environmental impact.^{3,32} All three CCD methodologies (A1 to A3) search for optimal solutions with respect to these system objectives.

The second input, (i2) *components pre-design*, identifies and proposes a first design of the relevant components and interconnections that comprise the system. The third input, (i3) *physics-based models*, defines the mathematical models that describe the dynamics of these components and interconnections, including high-, mid-, or low-fidelity models for mechanical and electrical subsystems, biological and chemical subsystems, control systems, economics, operation and maintenance, installation, materials, etc.

The fourth input, (i4) *real-data*, deals with variables and parameters measured at some lab or full-scale experiments, in real-time or offline, to facilitate the analysis of the system, validate the physics-based models, and reduced the inherent model uncertainty. The fifth input of the CCD process, (i5) *case studies*, chooses the internal parameters of the models and the external inputs that could affect the system, including operational cases, extreme load cases, system states, transition between system states and cases defined by the standards.

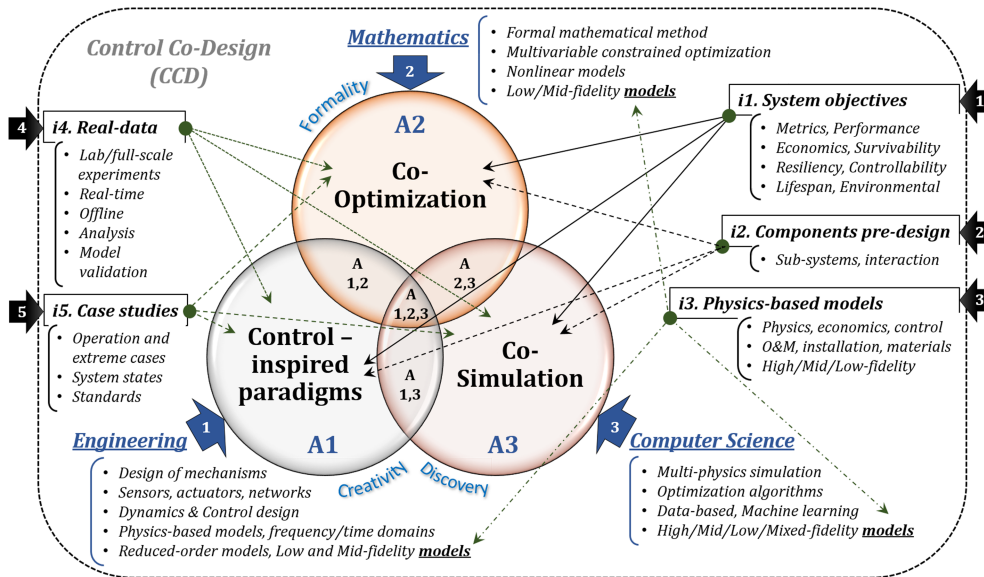


FIGURE 6 Control Co-Design areas: control-inspired paradigms, co-optimization, co-simulation

With the information given by these five inputs, the first CCD area, known as *control-inspired paradigms* (A1), proposes new design solutions based on a practical engineering understanding of dynamics and control. As discussed in Section 2, this was the approach applied by the Wright brothers and Charles Brush for the world's first heavier-than-air powered airplane and the world's first automatically operating wind turbine for electricity generation. Based on control engineering principles, this approach uses physics-based low-fidelity models in frequency and time domains, and tools like Bode and Nichols diagrams and Root-locus. It proposes new mechanisms, networks, and control solutions by looking at the fundamental control properties of the system, which typically include controllability, stability, disturbance rejection, reference tracking, resonance modes, damping, rise time, settling time, bandwidth, nonminimum phase characteristics, time delay, multiple-input-multiple-output coupling, integrity, fragility, observability, actuator-sensor colocation, control authority, nonlinearities, saturation, noise, and others.^{3,13,19,20,24}

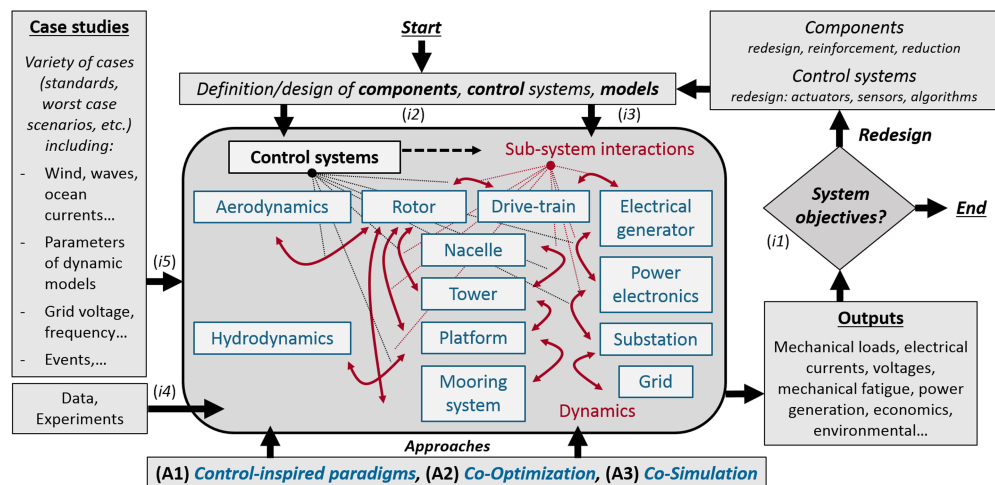
The second area, known as *co-optimization* (A2), uses a formal mathematical methodology with nonlinear low/mid-fidelity models and multivariable constrained optimization theory. The late 1980s/early 1990s projects in this area proposed a simultaneous optimization of the parameters of the controller and the plant, both with a fixed structure. This methodology was originally known as integrated structure/control design and was applied to airplane structures, mechanisms and spacecraft.⁴⁻⁸ They used mathematical programming approaches based on multiobjective constrained mixed-integer nonlinear optimization. The next generation of CCD co-optimization techniques went beyond these fixed-structure controllers and plants. They opened the possibility to change these structures during the optimization process. They proposed a global optimization exercise where the plant architecture, plant dynamics, and controller design are incorporated into a cost function within a nested-iterative optimization process with constraints^{16,33} or a multidisciplinary feasible architecture methodology.²⁵

The third area, known as *co-simulation* (A3), uses multiscale, multiphysics, high/mixed-fidelity dynamic models in an iterative simulation process.³⁴ In some cases, the simulation also includes optimization algorithms, data-based models, and machine learning techniques; see also Figure 6.

In all three areas, the accuracy and complexity of the multiscale, multiphysics, nonlinear dynamic models of the system are critical. The always-present uncertainty in the models of the system can complicate the design. The combination of parameters and potential inputs, and especially the inherent model uncertainty, makes the number of cases to study extremely large and computational prohibited. Monte Carlo techniques to select the appropriate parameter values and inputs can help the design process. Lab and full-scale experiments can also help to reduce the model uncertainty and facilitate the design process. This idea has materialized into two new paradigms. The first one, known as *Digital Twins*, combines the area (A3) co-simulation, with the input (i4) real-data, with real-time measurements from sensors distributed across the plant.³⁵ The second one, known as *experimentally infused optimization*, combines the area (A2) co-optimization, with the input (i4) real-data, with offline measurements from lab-scale experiments introduced in the middle of the design iterations to reduce the uncertainty space and help the optimization algorithms.²³

All three approaches present some benefits and some limitations. In principle, the *co-optimization* approach (A2) is able to find global optimum solutions for the system. However, quite often, it is not able to model all the complex aspects of the

FIGURE 7 Control Co-Design of a floating offshore wind turbine combining the three areas, ie control-inspired paradigms, co-optimization, and co-simulation, and including the five inputs: (i1) to (i5)



system or it fails to converge into a practical solution. Probably, the originality and initial success of the co-optimization approach at the end of the 1980s and mid-1990s narrowed down the CCD to only this area (A2), inhibiting CCD from being more extensively applied.

In addition, the *co-simulation* approach (A3), combined with data-based models and machine learning optimization techniques, is a very promising methodology for some systems. Note, however, that this approach is good for only some aspects of the design, as it shows significant limitations for finding creative solutions outside the information contained in the original data.

Lastly, the *control-inspired paradigms* approach (A1) is often able to keep an engineering understanding of the system and components during the design process, especially when appropriate control engineering tools are applied; see, for instance, QFT control design techniques.³ Studying the dynamic interaction among the components, this approach can propose new mechanisms, actuators, sensors, and control solutions to modify the system dynamics to achieve the design objectives. Combined with the other two areas, (A2) and (A3), it can suggest new practical solutions that are not achievable otherwise.

All in all, the intersections of areas in Figure 6, either two-area intersections (A1-2, A1-3, A2-3) or the three-area intersection (A1-2-3), are very promising approaches, opening the way to radically new optimal designs.

As a final example, Figure 7 combines the three areas (A1-2-3) and shows the five inputs (i1-i5) for the previous FOWT case. Starting with a first definition of the components and controllers (rotor, tower, floating platform, mooring system, control systems, etc), and for a number of critical inputs (wind, waves, grid conditions, etc), the method conducts numerical simulation campaigns that calculate the system outputs (mechanical loads, electrical voltages and currents, power generation, etc). These outputs are then compared to the system objectives (metrics). If these optimization objectives are not met, the process continues redesigning some components or controllers, restarting a new iteration. The iterations finish when the optimization objectives are achieved.

The three CCD areas need a concurrent engineering effort, involving engineers from different disciplines to redesign the components and control solutions at each iteration. Going beyond the complexity of the mathematical co-optimization techniques, or the computational limitations of the co-simulation campaigns, the combination of the three CCD areas, especially with the addition of control-inspired paradigms, enhances engineering creativity and opens the door to radically new optimal designs.

5 | CONCLUSIONS

This paper discussed the design process applied in industry to develop new products and systems. The paper introduced a concurrent control engineering strategy, known as CCD, which considers the multidisciplinary subsystem dynamic interactions from the very beginning of the design process. It combines the creativity of control-inspired paradigms, the formality of co-optimization techniques and the discovery of co-simulation methods to design the entire system and reach optimal solutions that are not achievable otherwise. The strategy is a *game changer* for the *control engineer*, who will be not only the designer of advanced control algorithms, but also the natural leader of the concurrent design of new products and engineering systems in industry.

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